

Engineering Decision Report

Project	MechAssist Engineering Analysis
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Date	2026-05-30 20:10
Revision	Rev 1.0
Generated	MechAssist v1.0

1. Engineering Narrative

Material selected: Steel SAE 9255 ($S_y=2.048$ GPa, concern: Fracture Risk). Stress: Safe at $\sigma_{vm}/S_y=0.013$ ($SF=74.29$, $target=2.0$, PASS, certainty 99.5%). Machining: Carbide at 53.9 m/min — tool life 3.16 hr, failure risk: Yes.

2. Material Selection — Module 1

Design Requirements

Min Yield Strength (MPa)	0
Max Density (kg/m3)	0
Min Elongation (%)	0
(Internal SI values)	$S_y=0$ MPa Density=0 kg/m3

Candidate Materials

Rank	Material	S_y (MPa)	Density (kg/m3)	Elong (%)	Failure Concern
#1	Steel SAE 9255	2048.0	7860	1.0	Fracture Risk
#2	Steel SAE 51440C	1896.0	8030	2.0	Fracture Risk
#3	Steel SAE 5160	1793.0	7860	4.0	Fracture Risk

Selection Rationale

Top-ranked material **Steel SAE 9255** selected based on highest yield strength (2048.0 MPa) within density and elongation constraints. Primary failure concern: Fracture Risk. Materials ranked by descending S_y , ascending density. S_u estimated as $1.3 \times S_y$ where tabulated value unavailable.

3. Stress Assessment — Module 2

Analysis Inputs

Member Type	Shaft
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Normal Stress s	0 MPa
Shear Stress t	15.92 MPa
Yield Strength Sy	2048 MPa
UTS Su	2662 MPa
Target Safety Factor	2.0
Allowable s_vm	1024 MPa (= Sy / SF_target)

Assessment Result

Safe

SF Check: PASS

SF = 74.293 vs target 2.0 | Certainty: 99.5% | s_vm/Sy: 0.0135

Von Mises Stress s_vm	27.57 MPa
Yield Strength Sy	2048 MPa
Allowable s_vm	1024 MPa
Safety Factor Sy/s_vm	74.293
Stress Ratio s_vm/Sy	0.0135
Model Certainty	99.5%

SF Check PASS: s_vm is 996.4 MPa below allowable (1024 MPa). Design margin satisfied.

Mohr's Circle

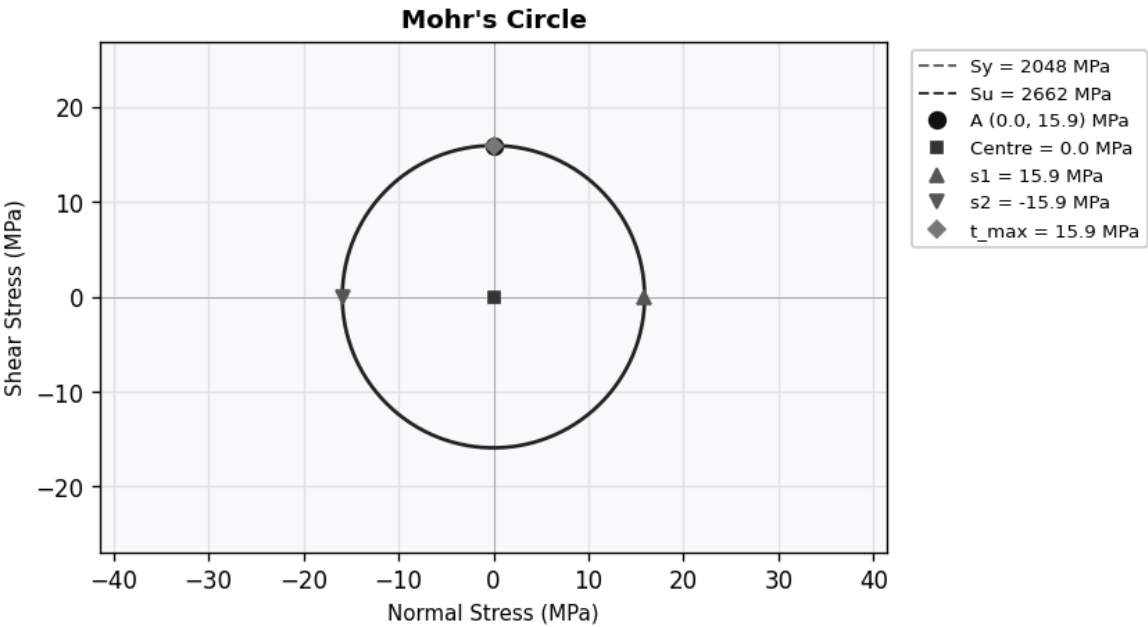


Figure 1 — Mohr's Circle: principal stresses and shear stress envelope.

4. Machinability Advisory — Module 3

Machining Parameters

Workpiece Grade	M
Tool Material	Carbide
Air Temperature	298.0 K
Process Temperature	580.0 K
Rotational Speed	343 rpm
Torque	6.2 Nm
Base Cutting Speed	53.9 m/min
Base Feed Rate	0.1 mm/rev
Base Depth of Cut	1.0 mm

Cutting Condition Advisory

Mode	Speed (m/min)	Feed (mm/rev)	Depth (mm)	Tool Life (min)	Failure Risk
Conservative	43.12	0.08	0.8	462.81	Yes
Balanced	53.9	0.1	1.0	189.57	Yes
Aggressive	64.68	0.12	1.2	91.42	Yes

Warnings & Recommendations

[Conservative] [!] Machine failure risk detected.

-> Reduce cutting speed or torque. Check spindle load and coolant flow.

[Balanced] [!] Machine failure risk detected.

-> Reduce cutting speed or torque. Check spindle load and coolant flow.

[Aggressive] [!] Machine failure risk detected.

-> Reduce cutting speed or torque. Check spindle load and coolant flow.

Tool Life Curve

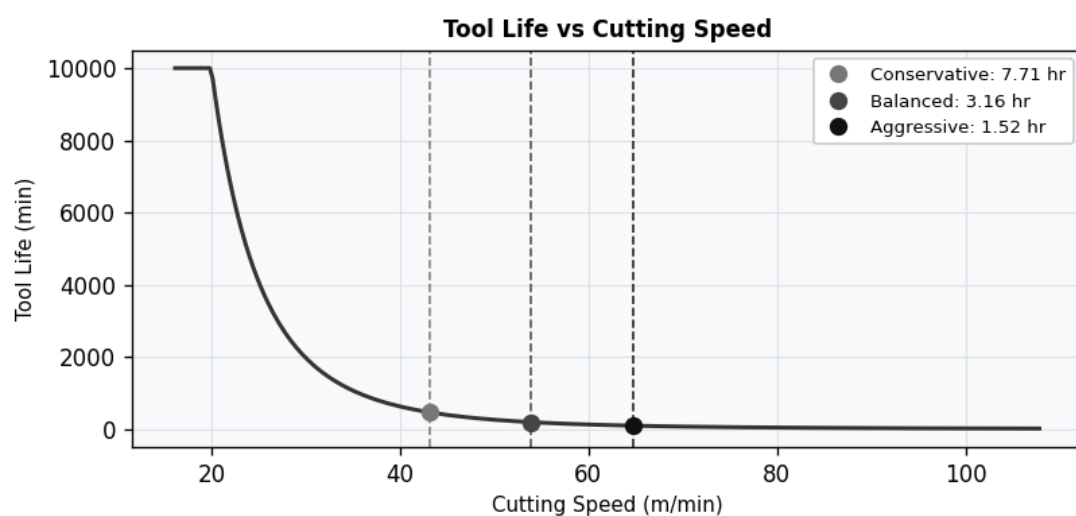


Figure 2 — Taylor tool life curve: tool life vs cutting speed for Conservative, Balanced, and Aggressive modes.

5. Conclusion

Selected material: **Steel SAE 9255** ($S_y=2048.0$ MPa, concern: Fracture Risk).

Stress verdict: **Safe**, $SF=74.293$ (SF Check vs target 2.0: **PASS**).

Recommended machining: Carbide tool at 53.9 m/min — tool life 189.57 min.

All results must be verified by a qualified mechanical engineer before implementation in a real design.

Disclaimer: This report is generated by an AI-assisted decision support tool. All results should be verified by a qualified mechanical engineer before implementation. MechAssist does not replace professional engineering judgment.